Pandas Data Analysis & Matplotlib Visualization

Python â¢ Unit 2: Data Analysis & Visualization â¢ Study Notes

DataFrame operations, group-by aggregations, time series, and plotting.

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1. Pandas Data Wrangling

Series and DataFrame indexing: loc vs iloc label/integer positional access.

Handling missing values: isna, fillna, dropna, and interpolation strategies.

Split-Apply-Combine paradigm with groupby and custom aggregation functions.

2. Visualization with Matplotlib & Seaborn

Figure and Axes object-oriented interface in Matplotlib.

Plotting statistical distributions: histograms, KDE plots, and boxplots.

Exporting publication-ready vector charts and heatmaps with colorbars.